

Simulation of Crop production function using AquaCrop-OSPy

This README file provides users with an overview of how we use AquaCrop-OSPy—an open-source Python implementation of the U.N. Food and Agriculture Organization (FAO) AquaCrop model—to simulate crop production functions in the Lake Diefenbaker area

Please note that these codes are designed to run in a Python environment. Ensure that you have a Python setup in place and that all required packages are installed before importing them.

Using virtual environment `'/Users/tharakajayalath/.virtualenvs/r-reticulate' ...`

```
# !pip install git+https://github.com/aquacropos/aquacrop
# from google.colab import output
# output.clear()
```

Now that AquaCrop is installed, we need to import its components into the environment.

```
from aquacrop import AquaCropModel, Soil, Crop, InitialWaterContent, FieldMngt, GroundWater
from aquacrop.utils import prepare_weather, get_filepath

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Running an AquaCrop-OSPy model requires selecting five key components:

1. Daily climate measurements
2. Soil selection
3. Crop selection

4. Initial water content
5. Simulation start and end dates

1. Climate measurements

AquaCrop-OSPy requires weather data for the cropping period being simulated. This includes a daily time series of minimum and maximum temperatures ($^{\circ}\text{C}$), precipitation (mm), and reference crop evapotranspiration (ET, mm).

2. Soil

Selecting a soil type for an AquaCrop-OSPy simulation is done via the Soil object. This object contains all the compositional and hydraulic properties required for the model. The simplest way to choose a soil is to use one of the built-in soil types provided in AquaCrop defaults. Users can modify any of these built-in soil types or create their own custom soil profiles.

3. Crop

The crop type used in the simulation is selected in a similar way via the Crop object. To define a crop, you must specify the crop type and planting date. Any of the built-in crop types (currently Maize, Wheat, Rice, and Potato) can be selected or user can define crop parameters. Users may also edit the built-in crop parameters or create custom crop definitions.

Broadly these crop parameters are divide into two categories:

- a). Conservative parameters
- b). Non-conservative parameters - Needs to calibrate based on local conditions

4. Initial water content

The initial soil-water content at the beginning of the simulation is specified using the *InitialWaterContent* object.

5. Simulation start and end dates

You must define the simulation's start and end dates (YYYY/MM/DD). These determine the period over which AquaCrop-OSPy will run.

Running the simulation

Additional model components may also be specified, including irrigation management, field management, groundwater conditions. Once you have defined the weather data, crop, soil, and initial water content, you are ready to run the simulation. To run the model, combine all components into an AquaCropModel object. This is also where you specify the simulation start and end dates.

AquaCrop-OSPy Simulation Configuration to Simulate Crop Yield–Water Relationship for Lake Diefenbaker